

Local Grand Canonical Monte Carlo Simulation Method for Confined Fluids

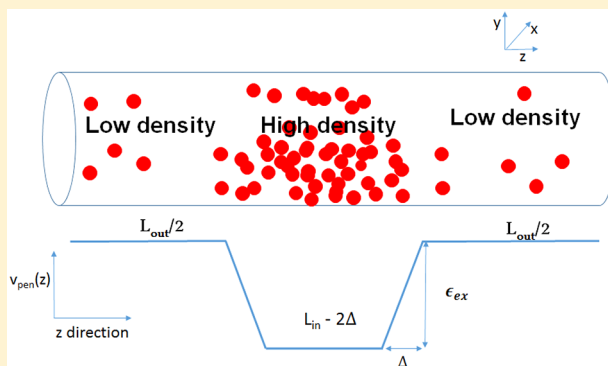
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ABSTRACT: We describe a new *local* grand canonical Monte Carlo method to treat fluids in pores in chemical equilibrium with a reference bulk. The method is applied to Lennard-Jones particles in pores of different geometry and is shown to be much more accurate and efficient than other techniques such as traditional grand canonical simulations or Widom's particle insertion method. It utilizes a penalty potential to create a gas phase, which is in equilibrium with a more dense liquid component in the pore. Grand canonical Monte Carlo moves are employed in the gas phase, and the system then maintains chemical equilibrium by "diffusion" of particles. This creates an interface, which means that the confined fluid needs to occupy a large enough volume so that this is not an issue. We also applied the method to confined charged fluids and show how it can be used to determine local electrostatic potentials in the confined fluid, which are properly referenced to the bulk. This precludes the need to determine the Donnan potential (which controls electrochemical equilibrium) explicitly. Prior approaches have used explicit bulk simulations to measure this potential difference, which are significantly costly from a computational point of view. One outcome of our analysis is that pores of finite cross-section create a potential difference with the bulk via a small but nonzero linear charge density, which diminishes as $\sim 1/\ln(L)$, where L is the pore length.



1. INTRODUCTION

Recently, supercapacitors, including electric double layer capacitors, have emerged as a promising alternative for traditional lithium ion batteries, due to their high power density, reasonably fast charging–discharging ability, and long cycle life.¹ However, their only moderate energy storage density remains a main limitation. To improve their performance on this front, ionic liquids (ILs) have been proposed for use as an electrolyte in order to increase the density of electrode screening charges, while porous materials are used as electrode material to boost the interfacial area and hence the amount of stored charge.² This combination of materials provides a unique challenge for researchers, seeking to understand the structure of ILs (fluid) at charged interfaces, due to the often severe confinement imposed by the porous electrode. Fluids under confinement often display properties quite different from the bulk, including the occurrence of surface or cavity driven phase transitions, which can give rise to anomalous behavior in electrical properties such as the electrode capacitance.^{3–5}

Theoretical investigations of such systems require a physically reasonable model for the IL and the porous environment. There are numerous studies that have introduced IL models ranging in complexity from those with polarizable, all-atom force fields^{6–9} to simple charged spheres.^{10–14} Within this broad spectrum we have also explored a number of simplified coarse-grained

models, as well as an off-center charged sphere model, which we have coined the anisotropic restricted primitive model (ARPM).^{15–17} Putting these models (even simple ones) into pores also presents certain technical difficulties. First, there is the choice of the geometry of the pore. It is most simply treated as a cavity with high symmetry, e.g., planar or cylindrical, which simplifies simulations or other theoretical approaches.^{18–20} Second, the confining surfaces of the pore will impose potentials on the fluid, e.g., hard walls, and dispersion or electrostatic forces,²¹ which makes the constrained highly coupled electrolyte a non-uniform fluid. In simulation studies, this means that maintaining chemical equilibrium may be problematic, using standard methods such as grand canonical Monte Carlo (GCMC),^{22,23} or thermodynamic integration.²⁴ In essence, the “holes” in a typical IL are rare, which makes the insertion steps difficult to carry out efficiently. The problem may potentially be further exacerbated in the relatively restricted confines of small charged pores, by the formation of “ordered” structures, more akin to one- and two-dimensional solids.^{4,5,25}

Many strategies, such as configuration bias Monte Carlo,^{26–28} continuous fractional component Monte Carlo,^{29,30} or weighted-histogram methods,³¹ have been developed to

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improve the efficiency of insertion moves. In addition, other techniques have also been developed that maintain equilibrium with an “unknown” bulk in non-uniform fluids.^{32,33} These latter works have been particularly useful, e.g., in benchmarking the accuracy of approximate theoretical methods, such as density functional theory.^{34–37} But the most pragmatic approach in recent times has been to simply simulate the IL in a finite pore, which is directly connected to the bulk liquid, with an a priori density.^{38–40} Such simulations rely on the assumption, that the correlation length in a typical IL is very short so edge effects created at the pore/bulk interface are small, allowing a good representation of the liquid in both the pore and bulk, while allowing free diffusion of particles between both environments. Simulation of a dense bulk is numerically very intensive, which is inefficient given that the major interest is the response of the fluid in the pore. This notwithstanding, this approach also solves another important problem for confined charged fluids, as it allows the direct measurement of the interfacial (or Donnan) potential that is required to maintain electrochemical equilibrium in such systems.⁴¹ It should also be pointed out that insertion methods that measure the *individual* ion chemical potentials can also be used to determine the relevant Donnan potential,⁴² but these suffer the same drawbacks as other insertion methods in dense fluids. The Donnan potential is crucial for the proper calculation of quantities, such as the electrode capacitance, that measure the response of the confined fluid while maintaining a constant potential in the bulk.

Here, we propose an alternative to the above approaches, wherein the weight of computational resources is concentrated back on the confined fluid. Put simply, we consider the fluid in the pore to be in contact, not with a bulk liquid, but with a gas. Furthermore, the gas phase need not be in a bulk environment, as the chemical potential can be easily measured either by insertion methods (such as Widom’s method⁴³) or else via GCMC. Using the GCMC route has the advantage of allowing one to maintain the chemical potential, while varying other properties in the confined environment. Such an approach can of course be applied to any type of fluid model, not just charged ones. However, for the case of charged fluids, the establishment of a well-defined Donnan equilibrium is still required, which we ensure by choosing a gas-phase environment which would maintain a constant potential difference with the corresponding bulk liquid.

Our method relies on establishing a gas/liquid interface by introducing a penalty potential, which creates a density difference between nominal *internal* and *external* regions of the pore. This approach shares some aspects in common with the so-called osmotic molecular dynamics method.^{44–46} In that case a pure fluid is modeled as a virtual two-component mixture of identical particles (discriminated by labeling). One of the components, the “solvent”, is allowed to pass through a semipermeable membrane, while the “solute” is restricted to a region of high density by the membrane. In our case, the penalty potential plays the role of a type of membrane but, since it acts on all particles equally, there is no need for differential labeling. Furthermore, the physical transparency of our approach allows us to more easily explore the role played by the shape of the penalty potential at the interface. This may be important in systems which show significant interfacial structuring or where particle transfer through the interface needs to be facilitated. In this context, we also note a similar device used by Powles et al. who applied a penalty potential to create regions of low density so as to obtain better statistics when using Widom’s method in

bulk fluids.⁴⁷ One can view our method as a generalization of this approach to the confined fluid and utilizing a localized version of GCMC simulations in order to fix the chemical potential, as we will describe below.

In this article, we test the method by first considering a simple Lennard-Jones fluid in different pore geometries. We then go on to consider a molten salt in a charged cylindrical pore, using a restricted primitive model fluid at high temperature and, as illustration, show how our method can be used to calculate the capacitance of a porous electrode for a range of pore sizes.

2. METHODOLOGY

2.1. Model. We begin by considering a fluid in a cylindrical pore. As shown in Figure 1, such a pore may be thought of as a

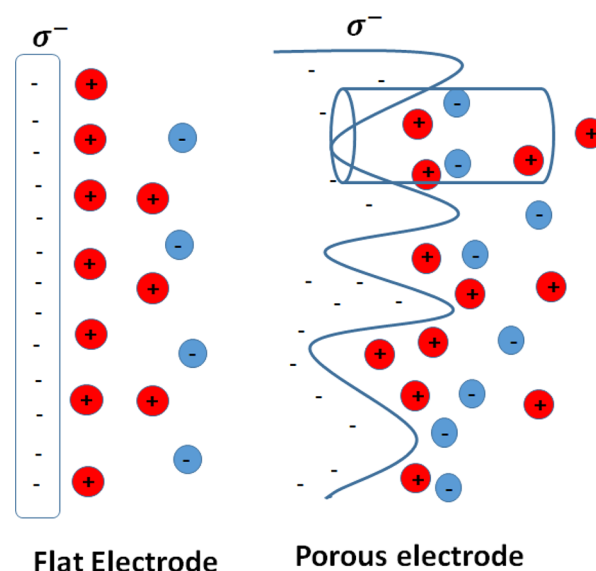


Figure 1. Simple cartoon showing an ionic liquid in contact with a flat and porous electrode (the latter is indicated by an undulating surface). We indicate how a pore may be approximated by a cylindrical volume.

representative model for a section of a macroscopic porous electrode. While depicted here with a circular cross-section, many different pore geometries may be encountered and, in this study, we will consider the effect of changing the cross-sectional shape of the pore. In all cases, the pore is assumed to consist of hard walls. That is, the surfaces of the cylinder, once defined (we shall describe the different shapes of cylinders we considered below), restricts the centers of all particles to be contained within its boundaries. This model ignores the role of dispersion interactions on the confined fluid properties, which are known, e.g., to have significant effects in ILs.^{21,48} However, the role of the surface interactions will not qualitatively affect the nature of our current investigation. Because the fluid in the pore is assumed to be in equilibrium with a bulk fluid, the modeling of the confined fluid should be carried out with constant chemical potential boundary conditions. If the fluid is charged, we must also account for the so-called Donnan potential, due to the charge distribution at the interface between the pore and the bulk, though, as we shall see below, in the case of a cylindrical pore, the interfacial charge profile (if it can be called as such) actually extends over macroscopic distances. However, we shall delay a discussion of the treatment of charged fluids until later in this article and choose instead to initially describe our method by applying it to a simpler fluid model. The extension to charged

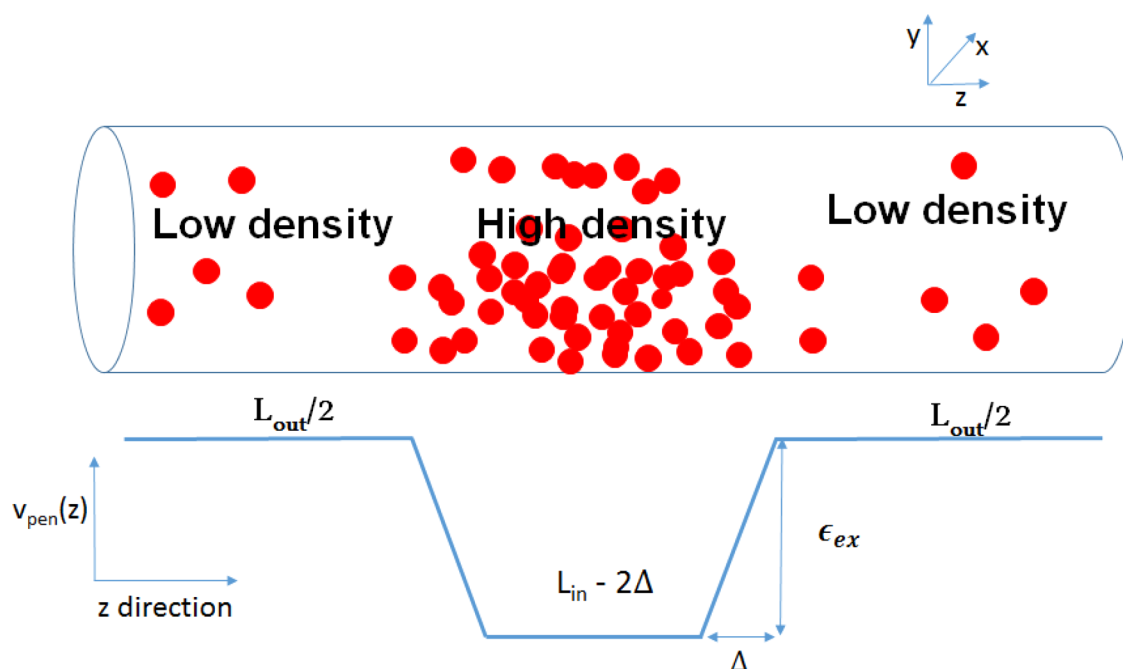


Figure 2. Diagram showing internal and external regions of the pore, created by the use of a penalty potential. The outer region consists of a gas-like fluid which is in equilibrium with the much denser liquid-like fluid in the inner region. Here we have depicted the *linear* penalty potential.

fluids is by and large fairly straightforward, though dealing with the Donnan potential does introduce a few subtle considerations.

Hence we consider a simple (neutral) fluid confined in an uncharged cylindrical pore, in equilibrium with its bulk. In a straightforward GCMC approach both insertions and deletions of particles would be carried out as part of a sequence of different simulation moves. Other types of moves, e.g., displacements, rotations, and cluster moves, etc., would be implemented accordingly so as to efficiently sample the important part of the configurational space of the model. We will consider a simple fluid of Lennard-Jones particles, which would require only displacement moves (in addition to insertions and deletions). As noted in the [Introduction](#), particle insertions become particularly problematic when the fluid density is high. The same is also true for techniques such as Widom's particle insertion method, often used for direct chemical potential evaluations.

Here we explore instead simulating a reference system whose chemical potential is relatively easily maintained, but which is able to exchange particles with the confined fluid via diffusion.

To this end, we divide the pore, with total length, L , into two regions. The inner region of interest contains the dense fluid, while the outer region consists of a dilute "gas" phase. The total length of the inner region is L_{in} and that of the outer region is L_{out} , such that $L = L_{\text{in}} + L_{\text{out}}$. This is schematically illustrated in [Figure 2](#). In order to maintain this density difference, we impose a *penalty potential*, which raises the relative chemical potential of the fluid in the outer region by a constant amount, ϵ_{ex} . Since it is the region of interest, we shall assume that the inner region extends over a volume which is large compared to the width of the interface between the dilute and dense phases. This guarantees that a substantial part of the dense phase mimics a fluid in an infinitely long pore. Particles will partition between both regions of the pore, while the chemical potential of the whole system can be maintained by carrying out GCMC simulations *in the outer region only*. Hence we will denote this approach as the *local-GCMC* (*l-GCMC*) simulation method.

Trials were carried out on two types of penalty potentials, which differed according to how the boundaries were treated. The first uses a tanh profile as follows,

$$\begin{aligned}
 v_{\text{pen}}(z) &= -\frac{\epsilon_{\text{ex}}}{2B} \left(\tanh \left(\lambda \left(z + \frac{L_{\text{in}} - \Delta}{2} \right) \right) + B \right) & -\frac{L_{\text{in}}}{2} \leq z \leq \Delta - \frac{L_{\text{in}}}{2} \\
 &= -\epsilon_{\text{ex}} & \Delta - \frac{L_{\text{in}}}{2} \leq z \leq \frac{L_{\text{in}}}{2} - \Delta \\
 &= -\frac{\epsilon_{\text{ex}}}{2B} \left(\tanh \left(\lambda \left(\frac{L_{\text{in}} - \Delta}{2} - z \right) \right) + B \right) & \frac{L_{\text{in}}}{2} - \Delta \leq z \leq \frac{L_{\text{in}}}{2} \\
 &= 0 & \text{elsewhere}
 \end{aligned} \tag{1}$$

where $B = \tanh(\lambda\Delta/2)$ and the z -axis is along the axial direction of the pore. The parameters Δ and λ control the width and

gradient of the potential at the interface. The second is a simple linear interpolation between the two regions; i.e.,

$$\begin{aligned}
v_{\text{pen}}(z) &= \frac{\epsilon_{\text{ex}}}{2\Delta} \left(z + \frac{L_{\text{in}}}{2} \right) - \frac{\epsilon_{\text{ex}}}{2} & -\frac{L_{\text{in}}}{2} - \Delta \leq z \leq \Delta - \frac{L_{\text{in}}}{2} \\
&= -\epsilon_{\text{ex}} & \Delta - \frac{L_{\text{in}}}{2} \leq z \leq \frac{L_{\text{in}}}{2} - \Delta \\
&= -\frac{\epsilon_{\text{ex}}}{2\Delta} \left(z - \frac{L_{\text{in}}}{2} \right) - \frac{\epsilon_{\text{ex}}}{2} & \frac{L_{\text{in}}}{2} - \Delta \leq z \leq \frac{L_{\text{in}}}{2} + \Delta \\
&= 0 & \text{elsewhere}
\end{aligned} \tag{2}$$

$$\mu = \mu_0 + \epsilon_{\text{ex}} \tag{5}$$

The linear penalty potential is depicted in Figure 2. As it turns out, our results proved to be rather insensitive to either choice for $v_{\text{pen}}(z)$, provided similar values were chosen for L_{in} and Δ . In particular, it is desirable to choose Δ to be at least several particle diameters, to dampen any structure at the interface as one proceeds to the interior part of the pore.

2.2. Simulations. Our simulations proceeded as follows. We used periodic boundary conditions in the z -direction, employing the usual minimum image convention in this coordinate. Particle displacements were performed throughout the simulation volume using the Metropolis algorithm⁴⁹ with a maximum displacement, Δr_{max} , that was the same for particles in the inner and outer regions. This was to ensure proper microscopic reversibility for displacements traversing the interface. The optimal value of Δr_{max} is limited by the higher density of the fluid in the interior, and this may lead to some inefficiency when sampling the configuration space of the outer reference fluid. However, this will be ameliorated to some extent by the use of insertion and deletion moves in the outer region. These serve to mix particles and, because they are efficient moves in a dilute fluid, they can be used relatively often. The specific definition of the “outer region”, where insertion and deletion steps will be allowed, is of course arbitrary, but in this work we chose it to be defined by the z -coordinate being in the range $L/2 > |z| > L_{\text{in}}/2 + \Delta$. In this outer region, particle displacements were coupled with the following GCMC steps for deletion and insertion. The probabilities of acceptance for these are given respectively by the following (usual) expressions,

$$P_{\text{del}} = \min \left\{ 1, \frac{N_{\text{out}}}{ZV_{\text{out}}} \exp \left(-\frac{\Delta U}{k_{\text{B}}T} \right) \right\} \tag{3}$$

$$P_{\text{ins}} = \min \left\{ 1, \frac{ZV_{\text{out}}}{N_{\text{out}} + 1} \exp \left(-\frac{\Delta U}{k_{\text{B}}T} \right) \right\} \tag{4}$$

where N_{out} is the instantaneous number of particles in the external region, with volume $V_{\text{out}} = (L_{\text{out}} - 2\Delta)A$, where A is the cross-sectional area of the cylinder. The quantity, $Z = \exp(\beta\mu_0)$ is the activity which is chosen a priori and ΔU is the change in energy associated with the insertion or deletion event. Here, $\beta = (1/k_{\text{B}}T)$ is the inverse thermal energy unit and μ_0 is the input chemical potential for the system. Therefore, as is the case with usual GCMC simulations, the activity Z defines the chemical potential of the fluid.

Within the inner region of the cylinder the penalty potential has the value $-\epsilon_{\text{ex}}$. The purpose of this potential is to create a fluid of relatively higher density in the inner region. That density difference is also affected by the choice of Z used in the outer l -GCMC simulations. We can define an *intrinsic* chemical potential for the dense fluid in the region, $|z| < L_{\text{in}}/2 - \Delta$,

The quantity μ is the chemical potential of the dense inner fluid were it isolated from the penalty potential. Thus, the l -GCMC method overcomes direct insertions and deletions into the dense interior of the pore by allowing particle exchanges through an open (but energetically biased) interface, but still allows us to evaluate the chemical potential of the isolated dense fluid directly. This leads to a much more efficient means for chemical potential evaluations in a dense confined fluid compared to direct insertion (or deletion) methods. However, the reduced numerical cost of this technique is accompanied by the undesirable introduction of an interface. It is possible that such an interface may give rise to unwanted structuring in the density profile (especially in the more dense fluid), which mimics oscillations seen for hard-core fluids against surfaces.⁵⁰ However, as we show below, for the fluid models considered here, it appears the length scale of the density variation at the interface is rather small (a few particle diameters). In those cases, where it may be necessary, it is possible that one may be able to reduce the effect of an inhomogeneous interface by manipulating the interfacial region of the penalty potential, a strategy that we did not need to apply in this work.

As part of our analysis, we also independently evaluated the chemical potential in various parts of the simulation volume using Widom’s particle insertion method. Widom’s method (WM) allows us to measure the chemical potential, according to

$$\begin{aligned}
\mu^{(\text{W})} &= \mu_{\text{ideal}} + \mu_{\text{ex}} \\
\beta\mu_{\text{ex}} &= -\ln \left\langle \exp \left(\frac{-\Delta U}{k_{\text{B}}T} \right) \right\rangle
\end{aligned} \tag{6}$$

Here, $\beta\mu_{\text{ideal}} = \ln(\rho\sigma^3)$ is the ideal part of the chemical potential, where ρ is the average particle density in the region sampled by the test particles and ΔU is the virtual change in potential energy associated with the trial insertion. The superscript in eq 6 indicates the WM evaluation. The bra-ket notation used, $\langle \dots \rangle$, denotes both an ensemble average and the average over the positions where the virtual insertions were made.

2.3. Simulation Parameters. As stated above, we tested our approach with simple Lennard-Jones (L-J) particles. In order to explore the influence of pore geometry, we considered circular, square, and hexagonal cross-sections for the cylinders. The sizes of the different cross-sections were defined by the linear scale R , which is equal to the radius for the circle, the side length of the square, and the apothem (perpendicular distance from the center to the edge) of the hexagon. The length scale is set by σ and energy by ϵ , which are the usual L-J interaction parameters,

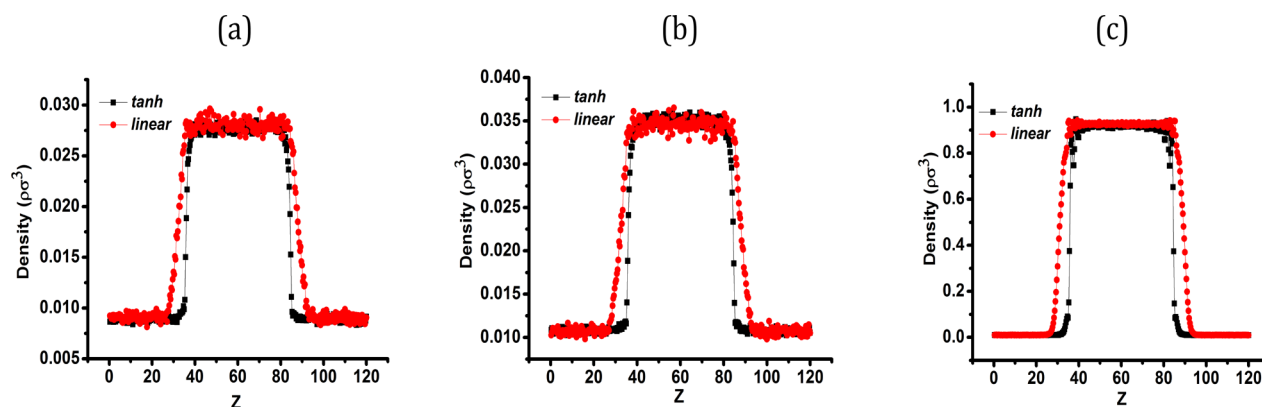


Figure 3. Comparison of the density profiles in the pore for various values of ϵ_{ex} and Z . From left to right the chosen parameter values were (a) $\ln Z = -4.64$, $\beta\epsilon_{\text{ex}} = 1$; (b) $\ln Z = -4.3$, $\beta\epsilon_{\text{ex}} = 1$; and (c) $\ln Z = -4.64$, $\beta\epsilon_{\text{ex}} = 4$. We show the results of using both the linear and tanh profiles for the penalty potential (with the same set of parameters).

$$v_{\text{L-J}}(r) = 4\epsilon \left\{ \left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right\} \quad (7)$$

The chosen values of ϵ and σ were not intended to mimic any particular species. However, a relatively large potential well, $\epsilon/k_B = 300$ K (k_B is Boltzmann's constant) was used, which provides a stringent test for the theory with the additional thought in mind that the theory will eventually be used to treat highly coupled IL models. We also used $\sigma = 2.3$ Å. The length of the cylinder was chosen to be $L = 52\sigma$, with $L_{\text{in}} = 28\sigma$ and $\Delta = 4\sigma$. The radius was $R = 2.5\sigma$, and the temperature, $T = 300$ K. For the range of moderate to high densities explored, a system of this size required between 300 and 1000 particles (approximately).

Most of the simulations were run for 8 million cycles, where in a cycle we attempt to move every particle once. Insertion and deletion steps in the outer region were attempted every 10 cycles and averages were calculated over the last 4 million cycles. We attempted 10 random Widom particle insertions every cycle, and subaverages for error determination were taken over every 1000 cycles. In some high density simulations though, where the Widom method is not particularly efficient, we increased the length of the simulations and the number of virtual insertions significantly in an attempt to obtain better statistics.

3. RESULTS AND DISCUSSION

3.1. Comparing Chemical Potentials. We investigated a number of different values for ϵ_{ex} and assessed the response of the L-J fluid in a cylindrical pore with a circular cross-section, with $R = 2.5\sigma$. To test the accuracy (and efficiency) of our method, we used the WM to obtain the intrinsic chemical potential by carrying out virtual insertions in an appropriately defined interior region wherein the z -dependent particle density profile, $n(z)$, is relatively flat. This was typically a few σ from the main interface. We shall denote the corresponding estimate for the intrinsic chemical potential using WM as, $\mu_{\text{in}}^{(\text{W})}$. The volume of this inner region, V_{inner} , was also used to calculate an average number of particles, $\langle N_{\text{inner}} \rangle$, and hence an average density, $\rho_{\text{inner}} = \langle N_{\text{inner}} \rangle / V_{\text{inner}}$, which then provides a reasonable representation of the liquid in an infinite cylinder. In some cases we also used the WM to calculate the chemical potential in the outer region as well.

The density profiles were obtained by sampling over bins distributed along the z -coordinate. Figure 3 gives typical results using both the tanh and linear penalty potentials, eqs 1 and 2.

These density profiles verify that, indeed, there is a relatively flat region of density in the central part of the pore, even for relatively high values of the interior density ($\rho_{\text{inner}}\sigma^3 \approx 0.9$). Furthermore, there is little difference between the results obtained for the two types of penalty potential, except for the expected variation in the interfacial region.

In Figure 4, we plot a comparison of the intrinsic chemical potentials obtained by the l -GCMC and WM as a function of the

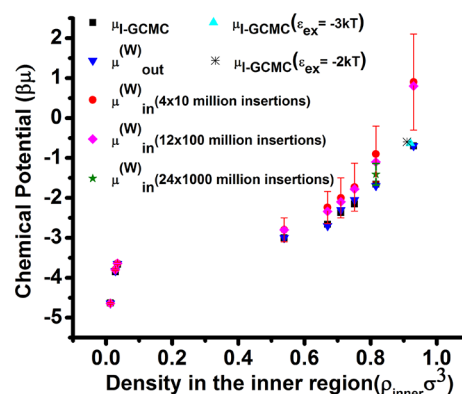


Figure 4. Intrinsic chemical potential (see definition in text) as a function of the inner density, ρ_{inner} , calculated using various approaches. The quantity $\mu_{l\text{-GCMC}}$ is the intrinsic chemical potential obtained using the l -GCMC method. At the highest density ($\rho_{\text{inner}}\sigma^3 \approx 0.9$), we used two different values for the penalty potential parameter, $\beta\epsilon_{\text{ex}} = -2, -3$ and adjusted the corresponding $\ln Z$, to give the same value for μ . The Widom method results were obtained either by insertions in the inner or outer regions, to obtain $\mu_{\text{in}}^{(\text{W})}$ and $\mu_{\text{out}}^{(\text{W})}$, respectively. In one case ($\rho_{\text{inner}}\sigma^3 \approx 0.82$), we performed a series of simulations of increasing length and number of trial insertions per cycle to calculate $\mu_{\text{in}}^{(\text{W})}$. These used 4 million cycles for equilibration and an increasing number of steps for calculation of the WM averages (4 million cycles $\times$ 10 insertions/cycle, 12 million cycles $\times$ 100 insertions/cycle, and 24 million cycles $\times$ 1000 insertions/cycle).

reduced inner particle density, $\rho_{\text{inner}}\sigma^3$. In order to check the veracity of the l -GCMC method, we simultaneously varied the values of ϵ_{ex} and Z , so as to maintain a constant μ , according to eq 5. Increasing Z leads to a larger density for the outer fluid, which, if compensated for by appropriately decreasing ϵ_{ex} , should lead to an invariant inner fluid density. These results are also plotted in Figure 4 (for the cases $\epsilon_{\text{ex}}/k_B T = -2$ and -3), which show excellent consistency in the l -GCMC result.

When the inner fluid density is not too large, the WM estimate, $\mu_{\text{in}}^{(W)}$, is close to that of the *l*-GCMC. We found that WM appears to be less reliable as the inner density becomes larger. We investigated how we could improve the accuracy of the WM evaluations for the case of a moderate reduced density ($\rho_{\text{inner}}\sigma^3 \approx 0.82$). To this end, we increased the length of the simulations to 16 million cycles (with 100 virtual insertions per cycle) and 28 million cycles (with 1000 virtual insertions per cycle). In all of these cases, the first 4 million cycles were used for equilibration. The results of these longer simulations are also given in Figure 4. In fact it is well-known that WM becomes problematic in dense fluids, due to the difficulty of finding holes in typical configurations of the dense fluid. In our case, this appears to lead to a systematic overestimation of the chemical potential using the WM. This phenomenon has been observed in bulk fluid simulations and has been attributed to an asymmetric distribution for the chemical potential evaluations at these high densities.⁵¹ Tellingly, each increase in simulation length gave WM estimates in closer agreement to the *l*-GCMC value, as well as an expected reduced estimated error. On the other hand, all of these simulations produced practically identical density profiles, which means there is essentially no change to the *l*-GCMC prediction for the chemical potential at the given density. Thus, one could safely argue that (not including the 4 million cycles for equilibrium) the *l*-GCMC method provides at least a 6-fold speed-up in chemical potential evaluations compared with WM, at the reduced density of ≈ 0.82 , and this would be significantly greater at even higher densities. Indeed, we would assert that the WM becomes essentially impractical for the observed “solid-like” structures we describe below. Similar comments apply when considering the relative merits of the *l*-GCMC vs *traditional* GCMC simulations, if the latter was used to simulate the high density confined fluid directly. For completeness, we also used the WM to evaluate the chemical potential using virtual insertions in the outer region, denoted as $\mu_{\text{out}}^{(W)}$. These were evaluated using the shortest simulations, but given the low density of the exterior region, they gave essentially perfect agreement with the *l*-GCMC values. We also used WM to evaluate the chemical potential in different parts of the interface between the dilute and dense regions for the profile with a reduced inner density of ≈ 0.82 ; see the inset in Figure 5. Here we obtained progressively worse estimates for the chemical potential as the density increased through the interface, in keeping with our findings above.

3.2. Fluid Phases in Pores of Different Geometries.

Figure 6 displays $\rho_{\text{inner}}\sigma^3$ vs $\beta\mu$ (from the *l*-GCMC) for different cylindrical cross-sections, with linear dimension, $R = 2.5\sigma$. The results are only shown for the linear penalty function. As we have already seen, our results are essentially identical for the two different types of penalty functions considered in this study. Two critical values of the chemical potential can be identified in Figure 6, characterized by a noticeable jump in the density (indicated by the arrows). The first occurs at $\beta\mu \approx -3$, where there is a large and relatively sudden density increase from $\rho_{\text{inner}}\sigma^3 < 0.1$ (“gas-phase”) to a “liquid-like” density, $\rho\sigma^3 \approx 0.6$.

The second jump occurs at $\beta\mu \approx 3-5$, which corresponds to a small increase in the average density, accompanied by a symmetry-breaking transition, reminiscent of a liquid to solid phase change. For convenience, we will label these two transition phenomena as *capillary condensation* and *crystallization*, respectively, though, because this system is inherently one-dimensional, true first-order phase transitions cannot strictly occur. Instead, the finite cross-section of the cylinders will cause

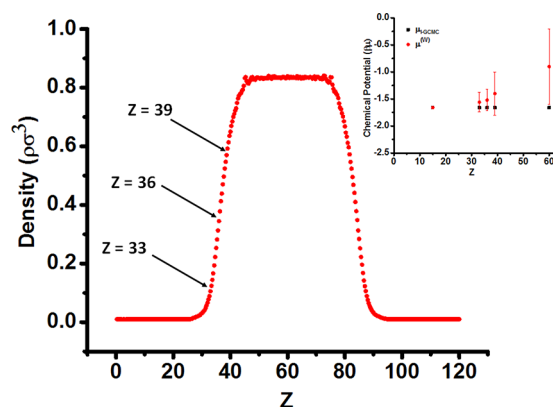


Figure 5. Positions at the interface which were sampled using WM for the case of a reduced inner density of ≈ 0.82 . The resultant chemical potentials are shown in the inset graph. The agreement with the *l*-GCMC results are excellent in the outer region (where the density is low) but become progressively worse as we proceed into the interface.

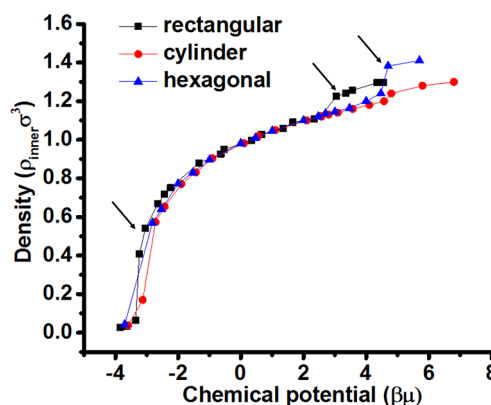


Figure 6. Plot of the inner density, $\rho_{\text{inner}}\sigma^3$, vs intrinsic chemical potential, μ , for pores of different geometry, with the pore linear scale, $R = 2.5\sigma$ (see text for definition). The arrows indicate the regions where there is a rapid increase in $\rho_{\text{inner}}\sigma^3$.

the condensation transition to be rounded within the pore. This is because the free energy difference between pure states (of equal free energy) and coexisting states with an interface will scale as γS , where S is the area of the interface (with minimum value equal to the cross-sectional area of the cylinder) and γ is the surface tension between phases. This free energy difference becomes infinite (signaling a proper first-order transition) as the surface area grows to infinity. These types of so-called capillary induced transitions have been well-studied in both cylindrical and slit-like pores, particularly in the seminal work by Gubbins and co-workers.^{52–55}

The apparent broken symmetry of the crystalline phase occurs because of the finite length of the inner part of the cylinder and will be *washed out* by concerted particle motions in a very long cylinder.⁵⁶ Recent fundamental work has identified the close-packed structures that can occur for hard and soft spheres confined in narrow circular cylinders, much smaller in cross-section than the pores studied in this work.^{25,57,58} Furthermore, we have not attempted to coerce the spheres into any regular structure, instead allowing the system to find “its own way” in configuration space, which means that structures that resemble close-packed forms may eventuate, but with inevitable imperfections. Strictly speaking, the definition of the inner region, as that where the density profile is relatively flat, cannot

be applied to the rapidly oscillating $n(z)$ of crystalline phases. We therefore defined the inner region for these systems to be the same as that for the fluid phases that immediately precede them (at lower densities).

The three different pore geometries displayed reasonably similar condensation values for μ but had noticeably different chemical potentials at which crystallization occurred. Crystallization happened at the lowest value of μ in the square pore and at greater values in the circular and hexagonal cylinders (the two latter cases were similar), reflecting the fact that the “templating effect” of the square cylinder is better able to induce ordered structures of Lennard-Jones particles, by imposition of a well-defined two-dimensional close-packed structure by the square walls.

Figure 7 shows the density profile, $n(z)$, in the inner portion of a square cylindrical pore overlaid on the chemical potential plot.

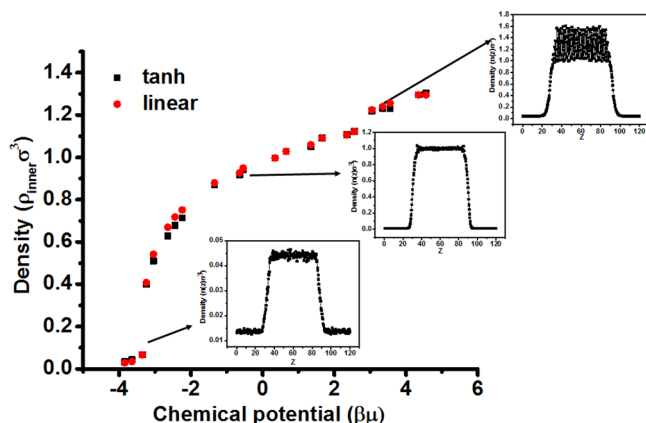


Figure 7. Plot of the inner density, $\rho_{\text{inner}}\sigma^3$, vs intrinsic chemical potential, μ , for a pore with a square cross-section with side length ($R=$) 2.5σ and typical density profiles in the gas, liquid, and crystal phases. Density vs chemical potential data derived from both types of penalty functions are shown.

For completeness, we show results for both types of penalty potential. We also plot a three-dimensional rendering of a typical particle configuration for the crystal in the square cylinder in Figure 8, and for comparison, we show a similar plot for the hexagonal pore. These configurations show clear ordering in the x – y plane as well as the z -direction, indicating that the crystal phases in these geometries consist of long ordered columns aligned along the pore axis. We should point out that the particles are rendered with a diameter smaller than σ so as to better visualize the structures formed.

In Figure 6 we saw that crystallization in the circular cylinder occurs at a higher value of μ than for the square cross-section. A typical crystalline-phase configuration in the circular cylinder is plotted in Figure 9. Here we see that the fluid forms three distinct cylindrical layers, presenting as concentric circles when projected on the x – y plane. We considered the outermost cylindrical layer, which lay within a relatively thin shell about a radial coordinate $\approx R$. This layer was *unwrapped* onto a two-dimensional plane by plotting $R\theta$ vs z (see Figure 9), where θ is the azimuthal angle. This plane projection shows an essentially triangular matrix of close-packed spheres, with imbedded imperfections. The wrapping into a cylinder, of close-packed spheres in two dimensions, has been used to describe the structure of close-packed spheres in narrow cylinders,^{25,57}

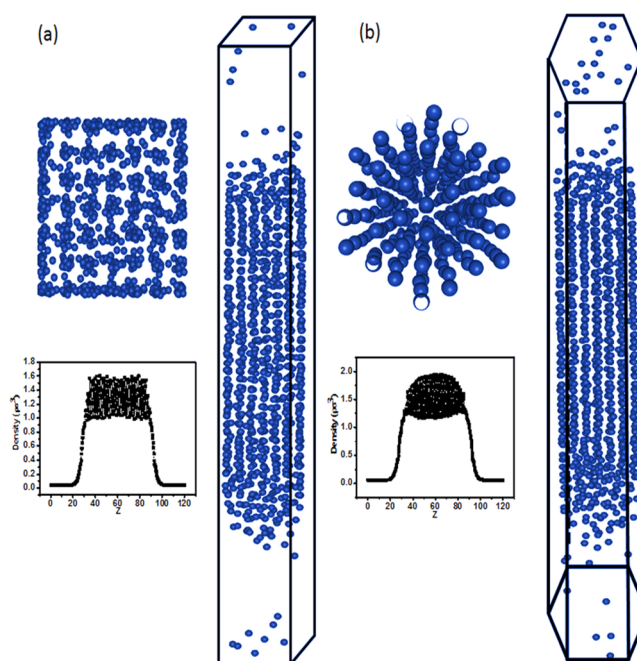


Figure 8. Rendering of typical configurations in the crystal phase for both the square and hexagonal pores with linear dimension, $R = 2.5\sigma$. See text for definition of the latter.

suggesting that the multilayered crystalline structure considered here may be described in a similar way.

Figure 10 shows the total radial distribution functions for the fluid in the different pore geometries in the liquid and solid phases. These distributions are qualitatively similar for the liquid phase in all pore types, displaying at least three distinct peaks. The first peak is strongest for the square cylinder, reflecting a generally larger degree of ordering in the presence of large flat surfaces. Upon the onset of crystallization, the radial distribution functions for the square and hexagonal pores display four noticeable peaks, reflecting the greater structure of the solid phases. On the other hand, despite the appearance of a pseudocrystalline phase for the circular cylinder, the extra structure is only marginally evident in the radial distribution function. In fact the latter appears to be very similar to that of the liquid phase, due to the much reduced order of the crystalline phase in the presence of the curved surfaces.

One motivation for studying different pore geometries is that porous electrodes in capacitor systems may well exhibit pore types with a range of shapes. Furthermore, there is some evidence that phase transitions induced in ionic liquids confined by pores may well have a significant effect on capacitor behavior.^{4,5} While geometry is one consideration, the cross-sectional dimension is another parameter which has been shown to be an important factor in determining capacitor performance.⁵⁹ We will consider the role played by the diameter in our studies below of a charged fluid in the presence of an electrified pore.

3.2.1. Charged Fluid in Cylindrical Pores. The model we choose to represent the charged fluid is the so-called *restricted primitive model* (RPM), which consists of hard spheres with a unit charge (either positive or negative) imbedded at the center of the sphere. We set the diameter of the particles to $\sigma = 5 \text{ \AA}$. The relative permittivity is equal to $\epsilon_r = 2$, and we assume a temperature of 1000 K, so as to prevent crystallization of the fluid, which mimics then a dense molten salt or ionic liquid (IL).

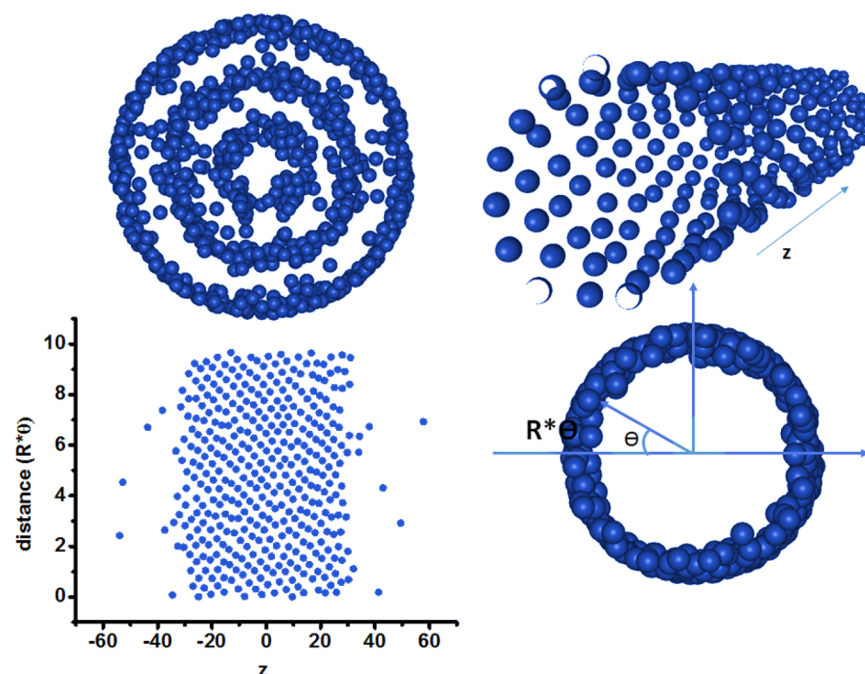


Figure 9. Rendered configuration of the crystal phase for a circular cylinder with radius $R = 2.5\sigma$. The inset plot shows the projection of the outermost layer (as depicted) on a plane. R^* is the radial coordinate for particles in the outermost layer with θ the azimuthal angle.

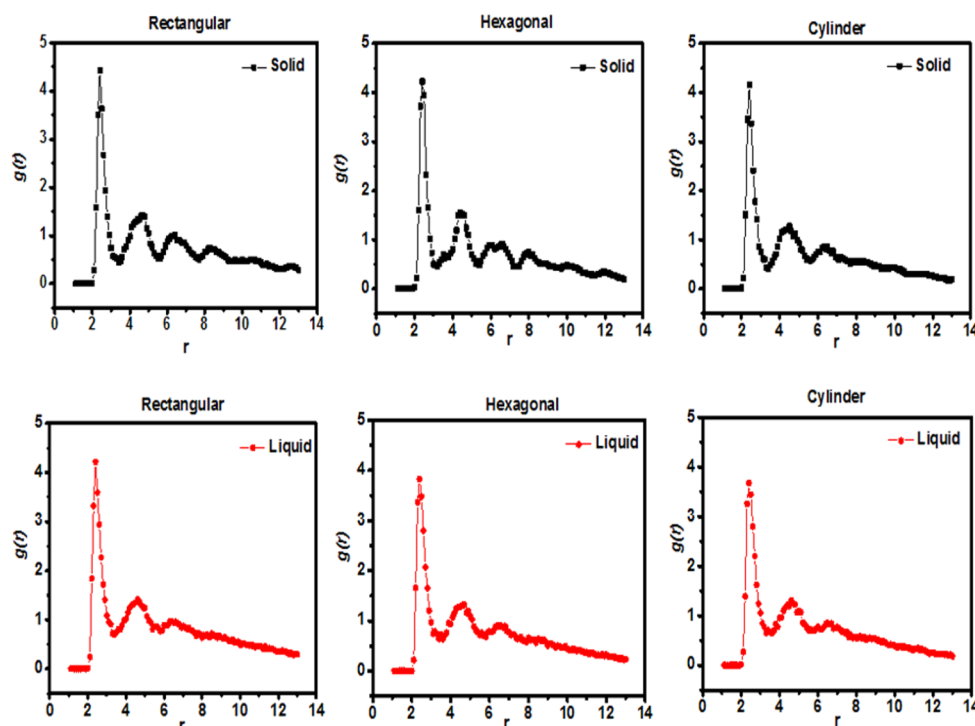


Figure 10. Radial distribution functions for the three different pore geometries with $R = 2.5\sigma$ in typical liquid and crystal configurations.

We note, however, that this model does not possess the asymmetry found in what are commonly known as room temperature ionic liquids (RTILs).

As stated in the [Introduction](#), there have been a number of studies in the literature that attempt to maintain the electrochemical potential of charged liquids in a pore by simulating an explicit bulk fluid that is in direct contact with the pore.^{38–40} Such simulations serve the dual purpose of fixing the mean input chemical potential, $\bar{\mu}_0$, as well as maintaining a fixed potential

reference in the bulk. For the 1:1 model used here, $\bar{\mu}_0 = 1/2(\mu_+ + \mu_-)$, where μ_+ and μ_- are the individual ion chemical potentials. Here, we will show how the *l*-GCMC method described above can be adapted to maintain constant electrochemical potential boundary conditions for confined charged fluids. That is, we will consider the IL in its gas phase in the outer region of the pore, which is in equilibrium with its liquid form in the inner region.

As before, a liquid is created in the inner region by applying a penalty potential; in this case we used the linear function, [eq 2](#),

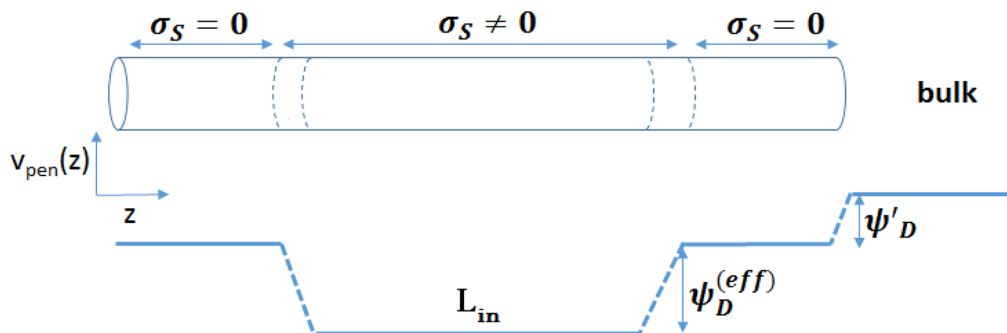


Figure 11. Cartoon depicting the different (charged and neutral) regions of the cylinder (and bulk) and the associated axial (and bulk) potentials.

with $L_{\text{in}} = 20\sigma$ and $\Delta = 10\sigma$. The outer fluid is maintained at a constant mean chemical potential by using l -GCMC simulations. As already noted, this method is efficient if the mean fugacity, $\bar{Z} = \exp(\beta\bar{\mu}_0)$, is small, consistent with a vapor phase. The l -GCMC simulations follow the same procedure as described above for the simple L-J fluid, except insertions and deletions occur for pairs of oppositely charged species. As simulations are initiated with a electro-neutral combination of counterions, co-ions, and surface charges, the total charge remains zero. The probability for accepting deletions and insertions, are respectively given by a generalization of our previous expressions; i.e.,

$$P_{\text{del}} = \min\left\{1, \frac{N_{\text{out}}^+ N_{\text{out}}^-}{(\bar{Z} V_{\text{out}})^2} \exp\left(-\frac{\Delta U}{k_B T}\right)\right\} \quad (8)$$

$$P_{\text{ins}} = \min\left\{1, \frac{(\bar{Z} V_{\text{out}})^2}{(N_{\text{out}}^+ + 1)(N_{\text{out}}^- + 1)} \exp\left(-\frac{\Delta U}{k_B T}\right)\right\} \quad (9)$$

where N_{out}^α is the number of $\alpha = +, -$ particles in V_{out} . The quantity ΔU is the energy change upon carrying out the insertion or deletion process. Particle displacements also follow the Metropolis algorithm as discussed earlier for the L-J particles. In some of our calculations the surface of the inner part of the pore, $|z| < \frac{L_{\text{in}}}{2}$, carries a certain charge (per unit area), σ_s . This surface charge is constructed by distributing discrete charged points evenly on the surface. This allowed us to calculate the particle–surface interactions by summing discrete Coulombic pair terms. We found that 1000 points gave good convergence to a continuous, uniform surface charge density and was more efficient than tabulating the bare electrostatic potential of the surface charge on the finite cylinder. We note that we did not take any surface polarization into account, implying a so-called ionophobic surface.⁶⁰ For particle–particle interactions we used the minimum image convention in the direction parallel to the cylindrical axis. This has the larger effect on the “gas” phase, as the Debye length in the concentrated liquid is very short. We can monitor the implication of this approximation on the density profiles, which, as we shall see below, remained well-behaved for all of the systems we studied.

3.2.2. Discussion on Electrochemical Equilibrium in Pores.

It is important to understand how our pore model is able to mimic the equilibrium that exists between a charged pore and a bulk IL. Such a scenario is analogous to the classic Donnan equilibrium between two electrolyte solutions separated by a semipermeable membrane. In that case, the two solutions have a difference in potential equal to the so-called Donnan potential. We can define a similar Donnan potential, ψ_D , in our cylindrical

geometry as the potential difference between the bulk and a point (or equipotential region) within the charged pore. Here we choose to define ψ_D as the potential (relative to the fixed bulk value) at the central axis well within the pore. This is a well-defined quantity for an infinite pore. For a finite pore, ψ_D would be defined in a region where the axial potential would be expected to be constant in some central section (for large enough L_{in}). So having defined the Donnan potential, the question remains as to how we can obtain ψ_D when we do not consider an explicit bulk, but instead the IL vapor.

To answer this, we note that if our system were placed in contact with a *dense* bulk fluid, we could create an equilibrium between that dense bulk and the IL vapor in the uncharged outer region of the pore. For example, this scenario can be created by using the same ϵ_{ex} in a penalty potential that acts between the vapor and the bulk. We can characterize the Donnan equilibrium between the vapor and the bulk with a second Donnan potential, ψ'_D , between the bulk and the potential along the central axis of the neutral pore; see Figure 11. Similar to the above, we must ensure that L_{out} is large enough to ensure this quantity is well-defined. Because we assume the pore which contains the vapor is uncharged, we will have $\psi'_D = 0$ for the symmetric RPM. For a more general *asymmetric* IL model $\psi'_D \neq 0$. That is, if anion and cation have a different chemical structure, which is typical in real ILs, even a neutral pore environment will give rise to an interfacial potential between the pore and the bulk fluid. For example, if a cation has a lower affinity to the neutral pore than the anion, due to non-electrostatic considerations, we would expect that the pore will have a negative potential relative to the bulk in order to bring about electro-neutrality deep within it. As we discuss below, this electro-neutrality is *local* in the sense that the total charge in any finite subvolume of the pore goes to zero as the length of the pore becomes infinite.

It is useful to define an *effective* Donnan potential, $\psi_D^{(\text{eff})} = \psi_D - \psi'_D$, which is the difference in the central axial potentials of the inner and outer regions of the pore. We reiterate that this assumes L_{in} and L_{out} are sufficiently long so that these potentials are well-defined and independent of the length of the simulated cylinder. In practice, the sufficiency of these lengths can be ascertained by sampling these potentials and ensuring there is a large enough region where they remain constant in the two sections of the pore. This can also be confirmed by monitoring the density profiles in these regions, which should be correspondingly flat. Properties of the confined fluid, such as the capacitance, require one to measure the change in pore surface potential, relative to a fixed bulk value, which is essentially equivalent to fixing ψ'_D instead. This latter constraint is most easily enforced by maintaining identical conditions in the outer region of the pore, while variations to the inner region

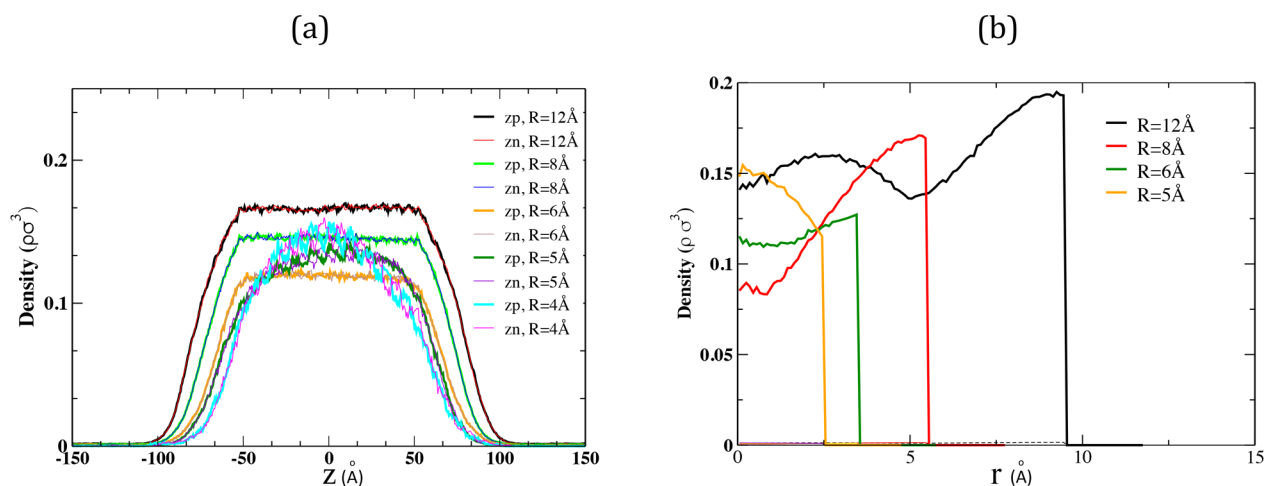


Figure 12. Density distribution of cations/anions for different pore sizes. *zp* and *zn* refer to cations and anions, respectively. Since the pores are neutral, the cations and anions have the same density profiles. (a) Density along axial (*z*) direction (Å); (b) density distribution along radial direction, *r* (Å).

(such as changing the surface charge) are carried out. In particular, the diameter of the neutral outer region of the pore should remain constant, as well as the fugacity in the *l*-GCMC simulations. On the other hand, for symmetric models such as the RPM, we will always have $\psi_D^{\text{eff}} = 0$ in a neutral outer pore with any radius (or geometry), which would afford us even greater flexibility in the modeling.

While the considerations above are reasonably self-evident, there still remains the question as to how the simulated system actually creates a constant potential difference, ψ_D^{eff} , between the inner and outer regions of the simulated pore. In the classic Donnan equilibrium scenario, a semipermeable membrane separates two half-spaces containing bulk phases with a charge separation occurring along the interface. When a permeating charge traverses the nonzero normal field at the membrane, it undergoes a potential jump which is maintained between the bulk regions, due to the essentially infinite area of the interface. Within each bulk phase, however, the solution remains strictly electro-neutral, because of the long range of the Coulomb interaction.

At the interface between the two cylindrical segments in our simulated system, there will also be variation in the charge distribution along the central axis as one traverses between the inner and outer regions of the pore. However, since the cross-section of the cylinder is finite, any charge separation of finite extent (in the *z*-direction) can be represented to lowest order as a dipole oriented along the axis. This would give rise to an electrostatic potential, which decreases to zero deep in the inner and outer regions of the cylinder (for large L_{in} and L_{out}). So an interfacial charge separation of finite extent cannot be the source of the potential difference, ψ_D^{eff} , between the pore regions. Instead, ψ_D^{eff} must be due to a *nonlocal* separation of charge distributed throughout both cylindrical segments. That is, a net linear charge distribution must exist in both regions of the pore and, furthermore, because ψ_D^{eff} must be finite, these linear densities must also go to zero (appropriately) as $L_{\text{in}}, L_{\text{out}} \rightarrow \infty$.

To determine the nature of this charge separation, we note first that a cylinder with finite cross-section and length, *L*, with a surface charge density of λ , will have an axial potential at its center of order $\sim \lambda \ln(L)$. This suggests that the linear charge densities in the inner and outer regions of the pore in our system will be of opposite sign (because the system is electro-neutral) and of order $\sim 1/\ln(L_{\text{in}})$ and $\sim 1/\ln(L_{\text{out}})$, respectively.

Interestingly, this means the total amount of charge that is separated in the two segments grows as $\sim L/\ln(L)$, in the limit that $L \rightarrow \infty$. This nonlocal excess charge gives rise to a fixed potential difference between the two segments but otherwise has essentially no direct effect (to order $\sim o(1/\ln(L))$) on the local fluid structure.

The intrinsic mean chemical potential of the fluid in the inner region is given by $\bar{\mu} = \bar{\mu}_0 - \epsilon_{\text{ex}}$ which can be adjusted by varying either (or both) $\bar{\mu}$ and ϵ_{ex} . In the studies presented here, the depth of the penalty potential was set at $\epsilon_{\text{ex}} = 2kT$ and the fugacity was chosen to give $\bar{\mu} = -29kT$. This gave an average density of approximately 3 M for the IL deep in the inner region of the pores. For the range of cylinder radii explored, this corresponded to between approximately 100 and 300 anion/cation pairs for neutral cylinders.

3.2.3. Particle Distributions in Neutral Pores. We investigated a range of pore radii, $R = 16, 12, 8, 6, 5$, and 4 Å. Recall that only pores with circular cross-sections were studied in these charged systems. The density profiles along the horizontal and radial directions are shown in Figure 12a,b. As the pore radius is reduced from initially large values, we see a decrease in density of the ions, seen in Figure 12a. This can be understood as being due to the excluding action of the hard surfaces, which will cause a reduction in the number of favorable interactions for those ions which are close to the surfaces. This effect becomes more pronounced if the curvatures of the surfaces increase (as the radius is decreased).

The reduction of average density is not monotonic, and for very narrow cylinders, $R < 6$ Å, the density at the center of the pore reverses trend and begins to increase as *R* decreases. This effect is related to the onset of apparent oscillations in the density profiles along the axial direction, which coincides with the ions forming a single file of alternating positive and negative charges, as the available space diminishes. The pronounced structure of this apparent one-dimensional crystal is really a consequence of the finite length of the inner region and will be smoothed by cooperative fluctuations as the length of the pore is increased. However, this strongly correlated arrangement of anions and cations promotes an increase in the average density. The apparent interfacial width of the density profiles appears to become larger for small pore radii. This effect is clearly seen for the cases $R = 5$ and 4 Å, where the oscillations in the density profile at the center of the pore are not regular and obviously

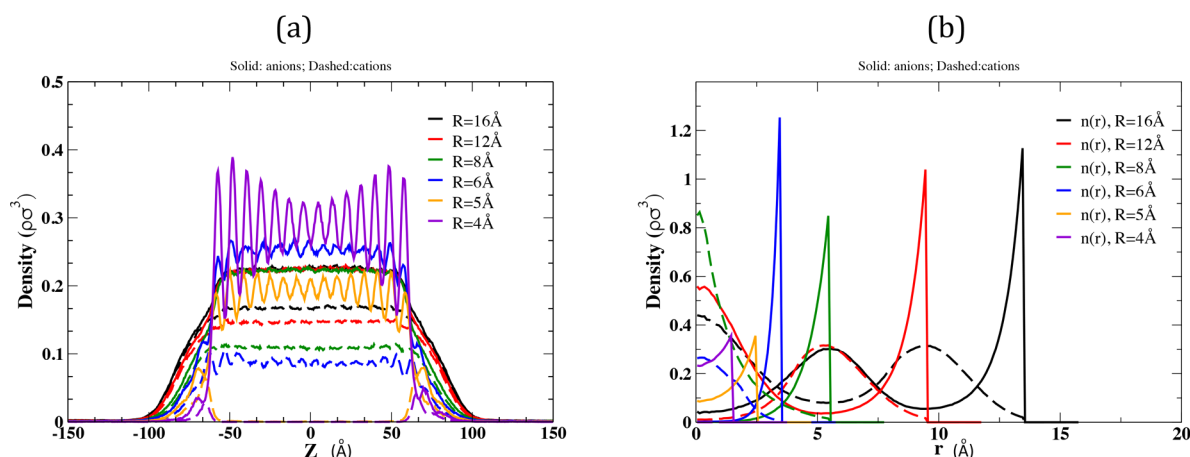


Figure 13. Density distribution of IL particles in charged pores of different sizes, with a constant surface charge density $\sigma_s \approx 0.0034e/\text{\AA}^2$. Cations and anions are represented by dashed and solid lines, respectively. (a) Density distribution along horizontal direction, z ; (b) density distribution along radial direction, r .

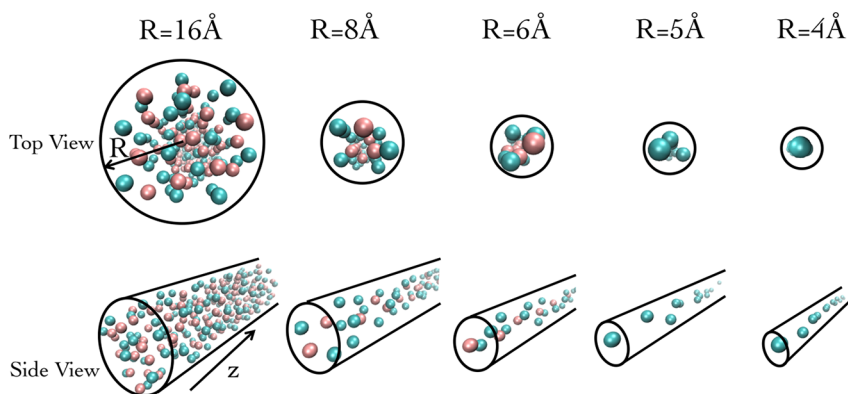


Figure 14. Rendering of anions and cations in a typical configuration in pores with different diameters. All pores have a surface charge density of $0.0034e/\text{\AA}^2$. Cyan and pink particles represent anions and cations, respectively. All configurations shown have liquid-like densities.

affected by the interface. This is a depletion effect driven by the low density of ions in the outer region of the pore, which again causes a diminishing of the attractive electrostatic interactions there. The density adjusts so as to reduce the corresponding surface tension, by creating a wider interfacial profile. While the same effect is present for pores with larger radii, fluctuations are larger in systems with a smaller cross-section due to a reduction in the overall surface free energy. While we did not explore this, the length L_{in} would need to be increased for smaller radii in order to obtain a significant region of the inner pore devoid of interfacial effects. Since narrow pores also have fewer absolute numbers of particles, such a strategy would be viable.

We also obtained the inner density distributions in the radial direction. Because the particle positions were measured at the center of the spheres, the maximum accessible value in radial direction is $R - \sigma/2$. These profiles, as shown in Figure 12b, show typical oscillations associated with particles packing at hard surfaces, tempered somewhat by the reduction in ion–ion interactions experienced near surfaces (as described above). The pattern of the oscillations depends upon pore size though, and for $R = 5$ Å there is only space for a single density peak at the center of the pore (reflecting the formation of a line of alternating charge), which also shows signs of depletion near the surfaces.

3.2.4. Particle Distributions in Charged Pores. We also investigated the effect of charging the pores. We considered a

moderate (positive) surface charge density of $\sigma_s \approx 0.0034e/\text{\AA}^2$ and investigated the effect on the density of varying the pore size, while the potential parameters and the chemical potential remained the same as those for the neutral cylinders. In Figure 13a,b, we once again depict the density profiles in the axial and in the radial directions. From Figure 13a, we see that reducing the size of the pores causes the difference, δn , between the average density of cations and anions to increase. As the radius (R) of the pore decreases for fixed (positive) surface charge density, the number of charges per unit length on the surfaces decreases as $\sigma_s R$. On the other hand, the number of charges per unit length in the electrolyte behaves as $\delta n R^2$, where δn is also the average charge density. Thus, the charge density in the screening electrolyte must increase linearly with inverse R in order to properly neutralize the surface charge. The results shown in Figure 13a indicate that for larger pore radii ($R > 6$ Å) this is largely achieved by lowering the cation densities in the pore as R decreases. That is, upon reducing the pore radius, the surface charge primarily acts to repel cations, while the anion density remains relatively constant. This can be understood as being due to the effect of electrostatic repulsion on the co-ionic species in a dense fluid environment, while the counterions remain strongly attracted to the surface charges.

With respect to the neutral pores, we see qualitatively different behavior once there is a noticeable structural component in the density profile ($R < 6$ Å). This is characterized by strong

correlations between anions and cations, and at $R = 6 \text{ \AA}$ the anion density increases slightly compared with the values at larger radii. At even smaller R , only anions are present in the charged region of the pore and their density increases as R decreases, until of course the radius is so small that ions are unable to fit into the pore at all. The pattern of removal of the ions becomes more apparent by investigating the radial densities (Figure 13b). Here we see that, as the pore radius is decreased, the outermost layer of counterions remains relatively intact, while internal layers of co-ions (and counterions) are sequentially removed. Figure 14 shows some typical configurations within the inner region, $|z| < L_{\text{in}}/2$.

3.2.5. Surface Potential and Integral Capacitance. We also measured the electrostatic potential at the surface of the pore, Ψ_s , as a function of the surface charge density, σ_s , for different pore radii. As discussed above, it was also necessary to obtain the potential difference along the central axis between the inner and outer regions of the pore, ψ_D^{eff} , in order to ensure that the surface potential is measured relative to a constant reference potential for varying σ_s . In this sense we can think of the outer region as serving as a “pseudobulk”, given that it maintains a constant potential difference, ψ_D , with an actual bulk. As discussed earlier, that potential difference is zero for the symmetric RPM, for all pore radii. For an asymmetric IL model the pore radius in the outer region would have an effect on ψ_D , and this must be accounted for when making potential measurements. As indicated by the flat density profiles in the central region of the pores (for the larger radii), the potential profile was reasonably constant over a sufficient range so that one could be sure that they represented the converged potentials for infinitely long cylinders. Where the density profiles showed significant structuring at small radii, the reported potentials were obtained by averaging over a chosen central region, though this of course led to more uncertainty. Once the potential difference along the central axis was determined, the surface potential could be obtained by considering the potential change in the radial direction from the axis to the surface. For each pore size, the (positive) surface charge was gradually increased so that the surface potential increase to around 1 V. Table 1 shows the

Table 1. Potential and Surface Charge Densities for Neutral (PZC) and Charged (CHG) Pores

$R/\text{\AA}^a$	$\sigma_s^{\text{CHG}}/(e/\text{\AA}^2)$	$\Psi_s^{\text{CHG}}/\text{V}$	$\Psi_s^{\text{PZC}}/\text{V}$	$\Delta\Psi_s/\text{V}$
4.0	6.121×10^{-3}	1.095	-1.056×10^{-2}	1.106
5.0	7.604×10^{-3}	1.201	2.897×10^{-3}	1.198
6.0	8.114×10^{-3}	1.158	8.296×10^{-4}	1.157
8.0	8.842×10^{-3}	0.983	1.678×10^{-5}	0.983
12.0	9.604×10^{-3}	1.100	1.629×10^{-4}	1.100
16.0	1.004×10^{-2}	1.152	2.210×10^{-4}	1.152
52.0	8.489×10^{-3}	0.984	0.000	0.984

^a R = pore radius.

measured potential at zero charge, Ψ_s^{PZC} , as well as the largest surface charge, σ_s^{CHG} , we considered, and its corresponding surface potential, Ψ_s^{CHG} , for each pore size. We note here that we expect the Ψ_s^{PZC} should be zero for our symmetric model. The fact that this is not always the case reveals the associated uncertainty in the potential, which ranges from between 10^{-2} and $<10^{-5} \text{ V}$, with the largest uncertainties occurring at the two smallest pore radii (as expected). From these, we calculated the integral capacitance, C_I , given by

$$C_I = \Delta\sigma_s / \Delta\Psi_s = \frac{\sigma_s^{\text{CHG}}}{\Psi_s^{\text{CHG}} - \Psi_s^{\text{PZC}}} \quad (10)$$

The corresponding C_I are plotted in Figure 15, which shows a marked decrease in the integral capacitance as the pore

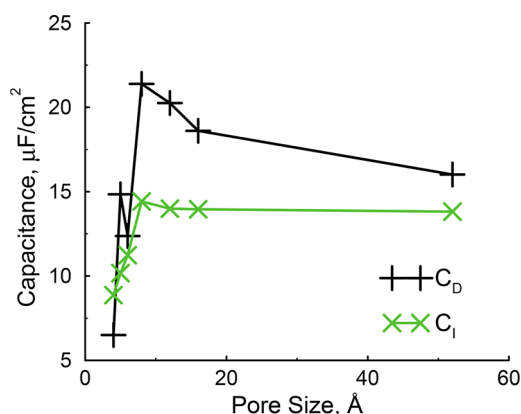


Figure 15. Variation of the integral capacitance and the differential capacitance (at zero surface potential) for different pore sizes. The black (plus signs) and green profiles (crosses) corresponding to the differential capacitance at zero potential and integral capacitance, respectively.

decreases in size below $R = 8 \text{ \AA}$. Just before that decrease, however, there is a minor increase in the C_I at slightly larger pore sizes. We also considered the differential capacitance, C_D , defined by

$$C_D = \partial\sigma_s / \partial\Psi_s \quad (11)$$

This was approximated by taking a finite derivative over incremental intervals, i.e.,

$$C_D(\Psi_{\text{av}}) \approx \frac{\sigma_s(i+1) - \sigma_s(i)}{\Psi_s(i+1) - \Psi_s(i)} \quad (12)$$

where $\Psi_{\text{av}} = 1/2(\Psi_s(i+1) + \Psi_s(i))$. The results for $C_D(\Psi_{\text{av}} \approx 0)$ are also plotted in Figure 15 and show that there is a significant increase in the differential capacitance for pore sizes ranging from about twice to equal to the ionic radii. This increased capacitance at small radius is related to the efficient packing of ions in the smaller pores, and the subsequent enhanced ability for counterions to effectively screen the surface charge density. Interestingly, experiments have shown that an increase in capacitance is expected as the pore size diminishes below roughly twice that of the ionic diameter, before it also decreases with even smaller pores.⁵⁹

4. CONCLUSION

We have developed a new *local* grand canonical MC method to treat fluids in pores in chemical equilibrium with a reference bulk. The method was tested using Lennard-Jones particles and shown to be much more accurate than direct insertion techniques such as Widom's method, implying that it is also superior to standard GCMC. There is some similarity between our method and a modified Widom method using a penalty potential by Powles et al.⁴⁷ and so-called osmotic MD,^{44–46} both used for bulk fluids. That similarity is further highlighted when considering charged systems where the Donnan potential is traditionally used to describe two electrolytes in osmotic equilibrium.

The main advantage of our method is that GCMC-type moves can be carried out in regions where they are efficient and the system then maintains overall chemical equilibrium by diffusion of particles into other regions. This way of varying particle numbers in the pore is similar in spirit to the method of putting the pore in direct contact with an explicit bulk simulation. However, a GCMC simulation of a liquid bulk would face issues similar to that of a dense fluid in the pore. In our method this problem is circumvented, as the reference fluid is a gas. The creation of an interface (in both these approaches) means that the confined fluid needs to exist in a large enough volume, so that a substantial part of it remains unaffected by the non-uniformity. In our approach, it is in principle possible to vary the profile of the penalty potential so as to minimize the interfacial inhomogeneity, though this was not deemed necessary for most of the systems considered here. Thus, it appears only a small number of additional particles need to be introduced (those in the interface and in the gas phase) when compared with a traditional GCMC simulation. It follows that there is no significant size effect in these simulations, provided one avoids systems where large interfaces may exist, e.g., close to critical points.

Finally, we have demonstrated the *l*-GCMC can find particular usefulness in simulating confined charged fluids. In these systems, one major issue is determining local electrostatic potentials in the confined fluid, which are properly referenced to the bulk. Previous approaches to this have used explicit bulk simulations (as described above). Alternatively, insertion methods that measure individual ion chemical potentials could also be used,⁴² though these would be problematic for such dense fluids. Using the simple RPM model for a molten salt, we have shown that sensibly referenced potentials naturally emerge from the *l*-GCMC approach, which can be used to obtain quantities of physical interest, such as the capacitance. One interesting outcome that emerged from this application was that equilibrium can only be maintained in pores of finite cross-section if they possess a small but nonzero linear charge density, which diminishes as $\sim 1/\ln(L)$, where L is the pore length. Because the *l*-GCMC method is particularly efficient, it can provide a useful way to assess the mechanisms for ionic liquid phenomena observed in experiments, such as capacitance. More sophisticated molecular models (compared with the simple RPM) will likely need to be employed in order to better mimic real systems. However, one of the advantages of our approach is that it is easily generalized to, e.g., many-center models for ionic liquids. The difficulties associated with performing insertion moves in simulations of dense fluids formed with oligomeric molecular models is well-known, and the *l*-GCMC method now emerges as a potentially important tool to deal with that problem. This notwithstanding, other theoretical techniques, such as density functional theory, have proven to be viable and even more efficient, theoretical approaches than simulations. Even so, fast simulation techniques, such as the one presented here, can serve to evaluate the accuracy of these more approximate theoretical approaches and provide strategically targeted benchmark calculations that can be used for their improvement.

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